ARP Spoofing Detection and Mitigation - Lab 3
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The goal of this lab was to create our own ARP spoofing detection tool. After building an ARP spoofing tool ourselves, we gained a better understanding of how the attack works. Using what we learned, we came up with a few ways to detect ARP spoofing effectively.
We brainstormed three main ideas for detecting ARP spoofing, but ended up implementing the first two since they were more practical and reliable:

1. **High Packet Rate Detection: We noticed that during an attack, the victim often gets flooded with ARP packets. So, we decided to monitor the rate at which ARP packets arrive. If the rate exceeds a set threshold, it's a red flag.**
2. **IP-MAC Mismatch Detection: Another common sign of an ARP spoofing attack is when the same IP address suddenly has multiple MAC addresses associated with it. If we detect that, it's a strong indicator that something suspicious is going on.**
3. **ARP Request-Response Imbalance: We also considered checking for an unusual number of "Is at" (ARP reply) messages compared to "Who has" (ARP request) messages. This could indicate that someone is constantly sending fake replies. (We didn't implement this one.)**

We divided the code into four main parts:

1. **Sniffing ARP Packets (packet_ARP): This function uses Scapy to sniff incoming ARP packets and passes each one to the detection functions.**
2. **Detecting IP-MAC Mismatch (noMatch): This function only deals with "Is at" messages. It checks whether the same IP is associated with different MAC addresses and, if so, raises an alert.**
3. **Monitoring Packet Rate (check_packet_rate): This function keeps track of how many ARP packets come from a particular IP within a short time. If the number of packets is too high, it triggers a warning.**
4. **Evaluating Attack Status (check_attack_status): After analyzing the packet, this function checks whether any of the detection flags were triggered. If yes, it prints a warning about the detected attack.**
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kali@kali: ~/PycharmProjects/PythonProject2

File Actions Edit View Help

```
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
No attack detected.
⚠ High packet rate detected from IP 10.0.0.138 - 31 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 32 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 33 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 34 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 35 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 36 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ Packet Flood attack detected!
```

kali@kali: ~/PycharmProjects/PythonProject2

File Actions Edit View Help

```
⚠ High packet rate detected from IP 10.0.0.138 - 77 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 77 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 77 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 77 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ High packet rate detected from IP 10.0.0.138 - 77 packets in 10 seconds
⚠ Packet Flood attack detected!
⚠ ARP Spoofing Detected! IP 10.0.0.138 has multiple MACs: 08:00:27:04:42:0f a
nd 00:b8:c2:c2:aa:59
⚠ High packet rate detected from IP 10.0.0.138 - 77 packets in 10 seconds
⚠🚨 You are under a combined ARP Spoofing and Packet Flood attack!
⚠ High packet rate detected from IP 10.0.0.138 - 78 packets in 10 seconds
⚠🚨 You are under a combined ARP Spoofing and Packet Flood attack!
⚠ High packet rate detected from IP 10.0.0.138 - 78 packets in 10 seconds
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```

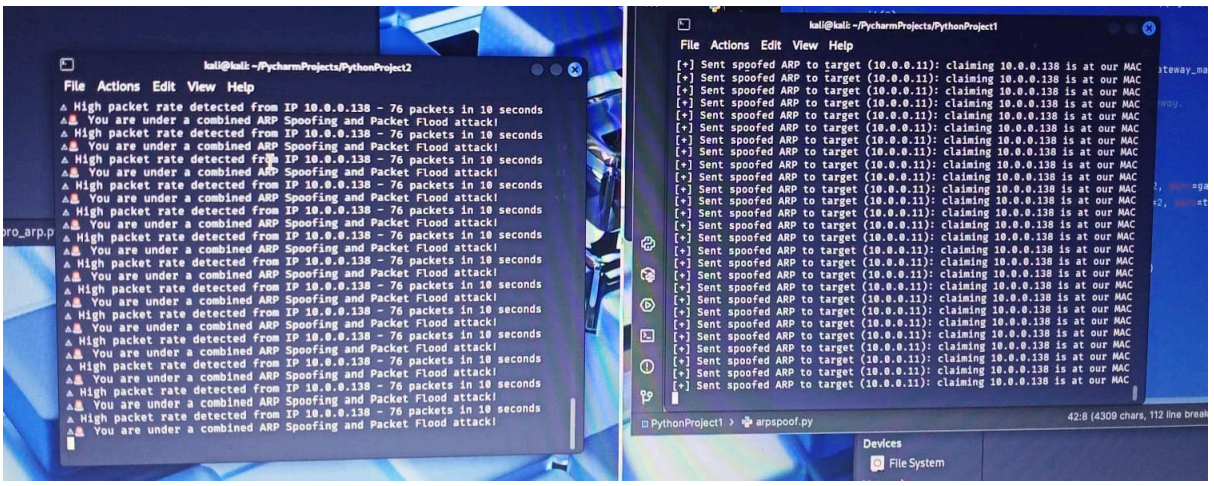

Our tool successfully detects ARP spoofing and packet floods in real-time. It works well in our testing environment, catching attacks based on both packet rate and IP-MAC mismatches.

bonuses

Detecting an attack is one thing, but stopping it is the real challenge. Here are some ideas we're considering to actually prevent or mitigate the attack:

- **ARP Packet Filtering: Dropping ARP packets that come from suspicious or known attacker IPs could help minimize the damage.**
- **Gratuitous ARP Responses: Once an attack is detected, broadcasting the correct IP-MAC mapping can help correct other hosts' ARP tables.**

we wrote a code that offers a way to actively mitigate the attack
in case we found an attack ->

```
def send_gratuitous_arp(ip_src, mac_src):
Send Gratuitous ARP to restore correct IP-MAC binding on the network
arp_response = ARP(op=2, psrc=ip_src, hwsrc=mac_src, pdst=ip_src,
hwdst="ff:ff:ff:ff:ff:ff")
send(arp_response, verbose=False)
print(f" Sent Gratuitous ARP to restore IP {ip_src} -> MAC {mac_src}.")
```